Using MsPASS for Data Processing on HPC and Cloud Systems

Session 1: Data Management with MsPASS

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MsPASS
Massive Parallel Analysis System for Seismologists

Introductions

- Gary Pavlis
  - 50-year history in computational seismology
  - Taught 35 years at Indiana University
  - Emeritus professor (old, “retired” guy) since 2018
  - Co-developer of MsPASS
  - I am doing all this as a volunteer in service to this community

Ian Wang



Career path:

- Ph.D. @Indiana University, 2017
- Postdoc @CREST, 2017 - 2018
- Research Associate @TACC, 2018 - now
- Manage the HPC P&A Group @TACC, 2022 – 2025
- Private industry since 2025

Coding learning path:

- Worked mostly in C++ during Ph.D.
- Picked up Python, Matlab, and others, too
- Learned parallel runtime systems during postdoc
- Seismology + HPC @TACC

Interest in Seismology:

- A seismologist since undergrad
- Mostly into signal processing and receiver functions during master
- Ph.D. work: 3D plane wave migration, 410 and 660 discontinuities, water in the transition zone
- Currently: seismic data management and processing, profiler, performance monitoring etc.

Course would not have been possible without Earthscope staff:

- Tammy Bravo
- Melissa Weber
- Gillian Haberli
- Sophia Parafina
- Chad Trabant
- Austin Ulakovich
- Interns: Asif Ashraf and Eshanta Mishra

Course Business

EarthScope Digital Credential

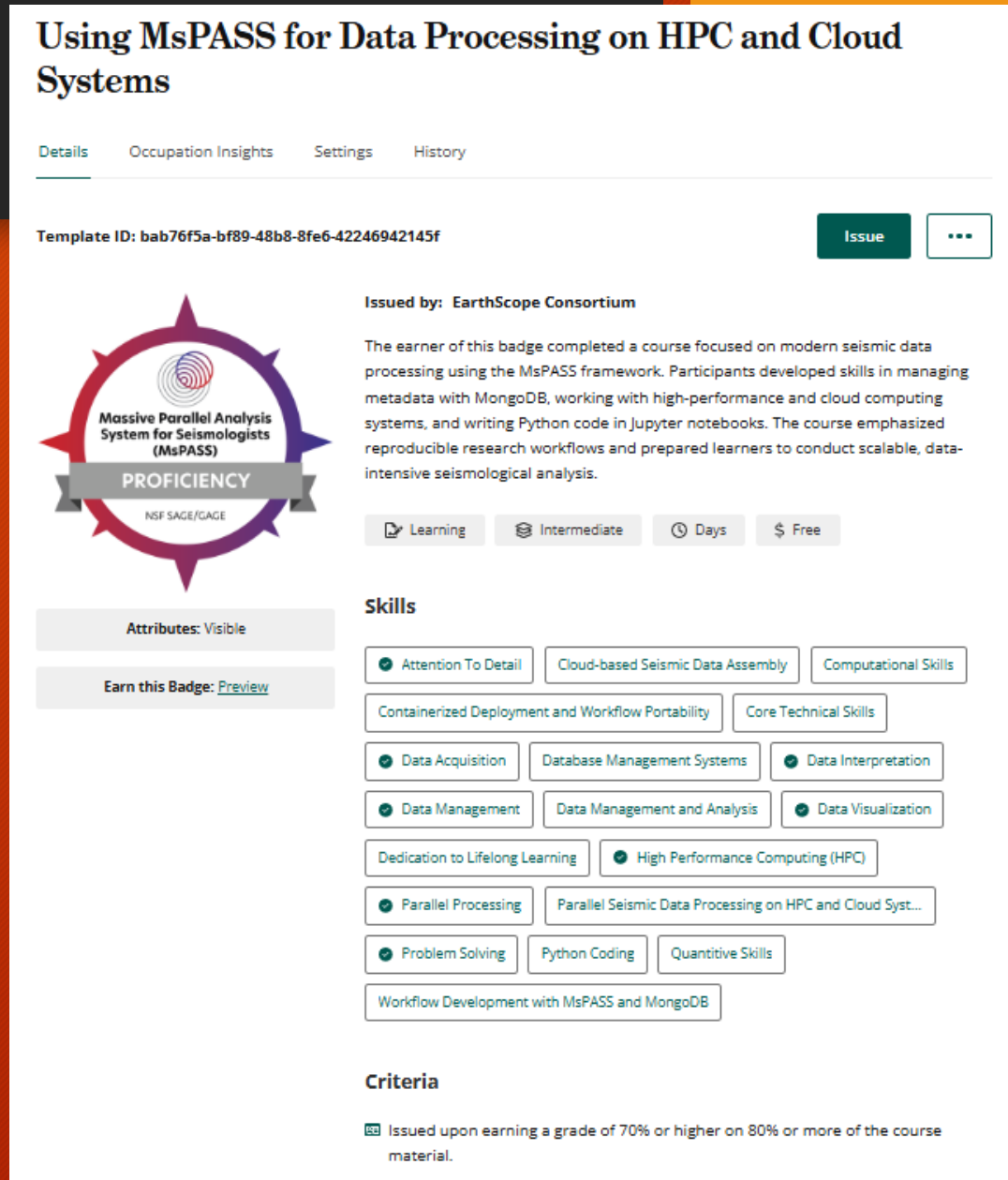
- By completing this course,
- you'll earn a Proficiency-Level Digital Credential

What it means:

- You've completed all required course activities, earning a grade of 70% or higher on 80% of the course material.
- You've demonstrated key skills through evaluated work
- Your achievement is verified and shareable

Why it matters:

- Showcases your skills on platforms like **LinkedIn** or on your **CV**
- Recognizes your applied learning and continued professional growth
- Becomes part of your professional learning pathway



How to Claim Your Credential

- **Step 1:** After the course ends, you'll get an email from **Credly**
- **Step 2:** Click the link in the email to **accept your badge**
- **Step 3:** Add it to your LinkedIn profile, resume, or digital portfolio
- **Have questions?**

  Email us at courses@earthscope.org



Axioms of the modern world:

- Everyone is an expert and everyone is an idiot
- Beware the Dunning-Kruger Effect (video title: "Why Stupid People Think They're Smart")
- Don't build your career on expertise in something that can be done by a robot

Career corollaries:

Expert

- To be successful you must become an expert in many things
- Know what you know and realize you know little else
- You must be a lifelong learner
- Continue to enroll in courses like this one

Idiot

- Be patient – remember you are one too
- Educate them, don’t run them down
- Preach and live the lesson of the Dunning-Kruger Effect

Career corollary 2: AI and the future

- Learn to use the AI as a tool in your work
- Avoid becoming an expert in areas the robot can do better
  - Encyclopedic professor – minutia
  - Problems dependent on busy work done better by a robot
- Develop problem solving skills
- Develop skills in assembling and interpreting information using the AI as a tool
- Learn to use tools like MsPASS that can exploit the new world
- Use the AI to hone your programming skills, not to do the work for you
  - Use AI guidance as a suggestion not gospel truth
  - An AI is a good teacher for the “how do I …” questions

Course Corollaries:

- Ask questions for all the rest of us idiots – highly likely you aren't alone
- Use good judgement because you may not know the difference
  - Your problem may or may not be unique
  - There are limits to what we can solve remotely
- Mechanism:
  - Zoom chat
  - Zoomp poll
  - Zoom breakouts if needed
  - Slack
  - Raise your hand if appropriate – stay muted until recognized
  - Attend office hour help sessions for personalized help

Assumptions

- You have listened to the precourse lectures
- You completed the precourse quiz and understand how to use Moodle
- You have run the precourse notebooks
  - Confirms you can get into GeoLab in the right configuration
  - Confirms you have a basic knowledge of jupyter lab
  - Confirms you have a working data set assembled for today's exercise

Assignments after each session

- Not just busy work
  - Designed to reinforce learning goals of sessions
  - Reading assignments important to improve learning for next session
  - You need to complete all of quizes for Earthscope certification
  - Completion requires only persistence
    - Wrong answers give feedback to improve understanding
    - Retry until you have all the answers right
- All done in Moodle
- Feedback important to keep us together
  - Questions/comment box for each assignment section
  - Attend office hours for personalized help
  - Will address any comments or questions in that box at the start of session2 and session3

Course Agenda

- Session 1: Big picture overview
  - Overview of MsPaSS
  - Data objects
  - Data management
  - Desktop setup and operation (an assignment)
- Session 2: Parallel processing with MsPASS
- Session 3: HPC clusters, cloud computing, and accessing the Earthscope waveform archives

How these sessions will work

- All sessions will begin with a short lecture overview
- Days 2 and 3 will have feedback review from homework
- Focus is hands-on work with Jupyter notebooks for each session
- Homework
  - Step 1 is always to read through the completed notebook text
  - Complete Moodle quiz

Using the Notebooks

- In the class we will lead you through the notebooks step-by-step and talking about what we are doing
- Questions first – details later
  - Answering questions is first priority
  - If we need to push forward, may skip some details
- Designed to be self-contained
  - You are expected to read the markdown box text
  - Pass 1 – skim during class sessions
  - Pass 2 – read again if needed as background for Moodle quiz
  - Follow hyperlinks and at least skim that material
  - If time runs short I'll give an overview and say "complete the rest of this after the session"
- For those unable to attend live zoom session
  - Listen to recorded lectures
  - Read through the notebooks running cells sequentially
  - Ignore the recording of the notebook sessions unless you get confused. Fast forward and/or ask a question on slack if you encounter problems

MsPASS

1. Massively Parallel
   - Single CPU thing of the past
   - Archaic software infrastructure

2. Analysis System for Seismology
   - Research analysis – flexibility
   - For Seismology research – tool for us not computer scientists

https://www.mspass.org/

A fundamental qualifier: “research computing”

Production versus Research IT systems

Production (Enterprise software)

- Solves Specific Problem
- Performance is critical
  - Time is money
  - Mission critical role with repeating tasks
- Operable with minimum skill set necessary for job – skilled labor is expensive
- Data model well known and fixed

Research

- Handle range of problems
- Performance is secondary
  - Being feasible sufficient
  - Many one-up solutions
- Can assume users are specialists and life-long learners
- Data highly variable (one person's signal is another's noise)

Key Roles MsPASS can play in research today

- The only game in town for novel ”big data” problems
  - Manage large data sets (terrabytes with millions of seismograms)
  - Parallel processing
  - Scalability allows prototyping on a desktop before running on big iron
  - Best tool today for accessing data from the Earthscope cloud waveform archive
- Tools for low-level work to assemble event-based data
- Generic parallel tools for handling continuous data
- Complete set of tools for P-wave receiver function processing
- Tools for spectral estimation
- Newer tool for automated processing of teleseismic P waves (son of dbxcor)
- Foundation for new implementation of our 3D migration code (pwmig)
  - MsPASS extension
  - Earthscope course scheduled for Novemeber

What is MsPASS good for

- Generic waveform processing toolkit
- Scalability
  - Prototype on a desktop
  - If needed scale up to HPC or cloud
  - Likely to become the tool of choice for interacting with Earthscope DMC when archive is completely migrated to the cloud
- Data management (integrated database)
- Reproducible science: With GeoLab your MsPASS notebook will allow reproducing at least the low-level processing to assemble your dataset

Aside: Reproducible Science

- Today a large fraction of seismology papers should be published in the *Journal of Irreproducible Results*
- Reasons:
  1. Data assembly sequence impossible to recreate
  2. Data processing sequence impossible to recreate
  3. Use of specialized research code not available to reader
- Solutions
  - 1 and 2 are solved by MsPASS
    - it was a design goal
    - The data set and notebooks used in this course are an example
  - Journals now universally address 3 - “Data” statements in more recent papers

MsPASS is NOT

- Ideal solution for problems we viewed as solved:
  - Real time data handling
  - Seismic event catalog processing
  - Seismic reflection processing
  - Archival data management for Earthscope and other data centers
- A complete solution for all of seismology
  - This is a framework, not a turnkey solution to all problems
  - May require some work to develop your custom “workflow”
  - Use of python as job control language makes flexibility feasible
  - More custom solutions require contributions from people like you
- A commercial product
  - 100% open source – free
  - Future depends on the community

What is in the box?

- Exceptionally complete documentation (https://mspass.org) for open-source package
  - Assume you have all had a quick look at that material
- Database to manage large data sets
- Parallel processing
  - HPC or cloud functionality
  - Clean error handling for long running jobs
- Flexible but simple to use IO abstractions (read-write almost any format and from URL)
- Python job control language
  - Reproducibility automatic if you submit jupyter notebook with your paper
- Algorithms – low level preprocessing (solved problem) focus
  - All obspy signal processing algorithms
  - Low-level windowing, bundling, and similar grungy stuff
  - Three-component primitives (rotation and transformation)
  - Trace editng
  - Header math
  - SNR calculations
  - All common P receiver function deconvolution
  - Array-based body wave arrival time estimation with multichannel correlation
  - Prototype array P wave deconvolution
- Docker container or anaconda package distribution

Sermonette/Sales Pitch

- MsPASS makes archaic
  - SAC
  - AH
  - Most, if not all, existing generic waveform handling tools
- MsPASS supercedes ObsPy
  - If you use ObsPy you can use MsPASS with minimal effort
  - MsPASS has integrated data management and parallel processing not possible with ObsPy
- MsPASS is now a rock-solid package
  - Processed many millions of seismograms
  - Haven’t seen a seg fault in 3 years
  - Most errors are user errors or unexpected data problems

I know ObsPy well, why should I move to MsPASS?

- MsPASS is obspy on steroids
  - MsPASS is a superset of obspy
  - All obspy waveform processing functions are part of MsPASS
  - Most obpsy python scripts are easily convert to a MsPASS script
- For large data sets
  - Integrated database for data management essential
  - Clean error handling to run large jobs running for hours
  - Parallel processing functionality
  - Cross-platform compatibility – runs on desktop to highest end supercomputer or cloud system
- Cloud computing capable
- Reproducibility
- Key thing: built for the future while obspy was designed almost 20 years ago

Documentation, documentation, documentation:

- User Manual
- Python API Reference
- C++ API
- Tutorials – you are all using files in a directory of this repository

IMPORTANT Consequence of good documentation

- The reference material is there if you know the layout:
  - The last slide was an overview
  - RTFM
- The current generation AI's are very good at answering questions about MsPASS
  - The reason they are good is because the user manual is large
  - I'll show you an example
  - Because of AI capabilies I will completely gloss over "implementation details" and focus on fundamentals

Data Management

- First a little online poll

Data Management: size matters

- For humans handling more than a few hundred “things” at once is problematic
- “Handling” means:
  - Being able to find the thing(s) you want to work with
  - Have the data organized into a stack of stuff in logical groupings
    - Paper files are the original database – centuries of experience prior to digital age
    - Computers files and folders (directories) are the logical extension of paper files
  - Being able to comprehend the contents of things you “find”
    - If you are an English speaker and a paper file is in Chinese you need a translator (data format)
    - For computer data you need to know the format and translate what the machine reads to something you can digest
- “Big data” means a Database system is required to manage information

Database Management Systems (DBMS)

- Unix file system and the hierarchic database
  - Evolved from fundamental concept of a tree structure in early computer science
  - Viewed as archaic in the IT world
  - Still used by large fraction of seismologists as unix directory file structures
    - Directory path is grouping (e.g. year/day directory like 2012/245)
    - Files (leaf of tree) are waveform data or other data files (name convention)
    - We will see the Earthscope archive has a legacy structure like that
- Relational Database systems
  - First used in seismology in nuclear monitoring research
  - Now a standard part of most network operations but “under the hood”
  - Relational == Tables
  - You likely use multiple RDBMS systems ”under the hood” every day when you ran many modern internet “Apps”
- Document Database
  - Used in MsPASS
  - Used because it is far less rigid than an RDBMS – more suitable for research applications
  - Unconventional, but
    - MongoDB is as stable as any RDBMS
    - You have likely used MongoDB when running some internet Apps. You can’t tell from the application laye

Poll question 2:

- Helpful for me to know my audience
- Do the poll while we take a short break
- After break we will work on Session1 notebook on GeoLab

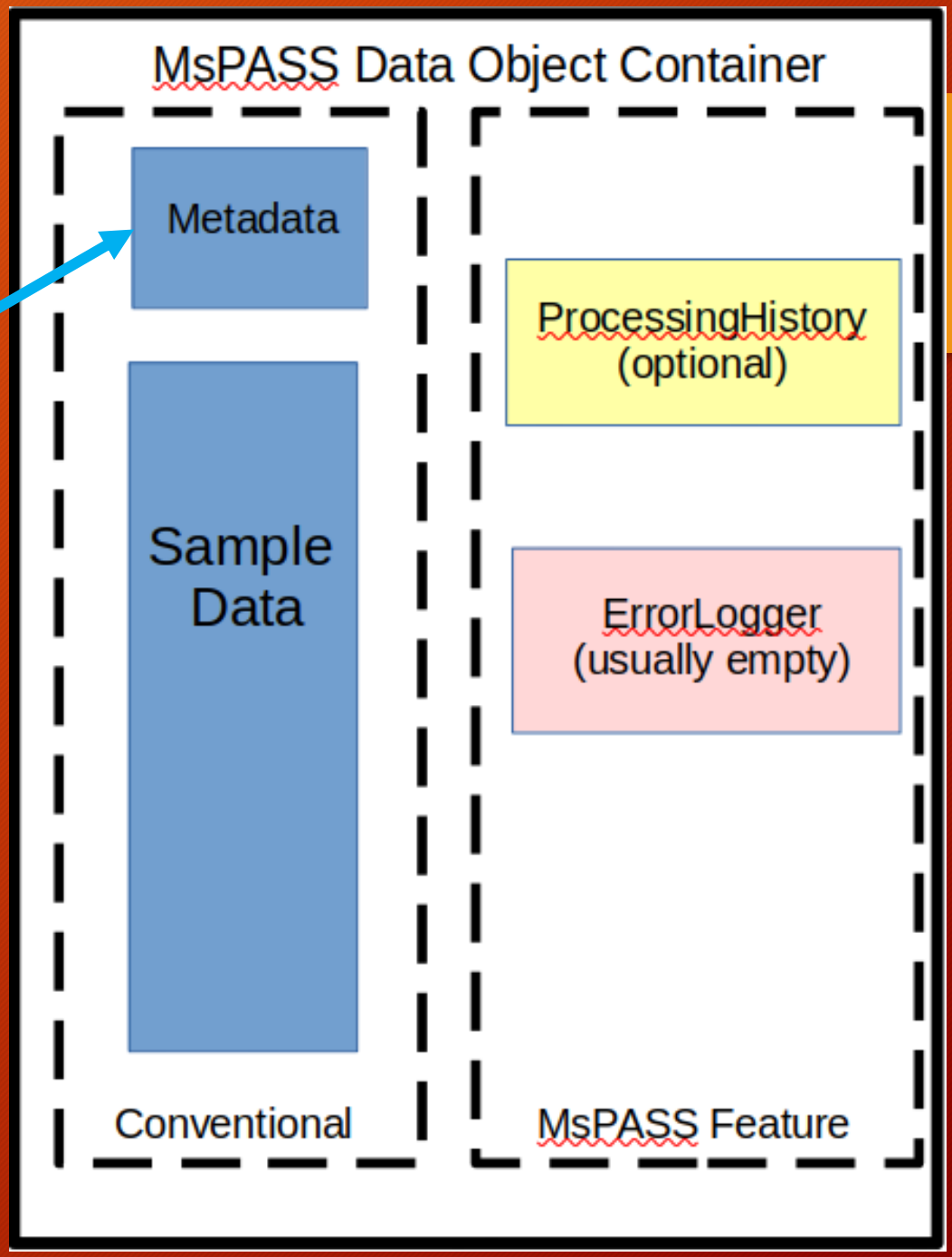
Review of Key Concepts of Part 1 of the notebook

MsPASS Data Object Concepts

- Metadata and sample data are different things and need to be handled differently
  - Sample data stored in “array-like” containers
  - Header data stored in a “dictionary-like” container (Metadata)
  - ”Is a” (MsPASS) versus “has a” (obpsy) perspective on header data
- For parallel processing we require an ”ErrorLogger” container

MongoDB collection

```
{ “npts” : 1024,
   “t0” : 601489897876697.89,
   “sta” : “AAK”
   ”chan” : “BHZ”
   …
}
{ “npts” : 1024,
   “t0” : 601489897876684.24,
   “sta” : “BZN”
   ”chan” : “BHZ”
   …
}
…
```



For geeky details see:
https://www.mspass.org/user_manual/data_object_design_concepts.html#

Part 2: MongoDB fundamentals

Generic Database concept: CRUD

- Create
  Stock MongoDB
  - Db.insert_one
  - Db.insert

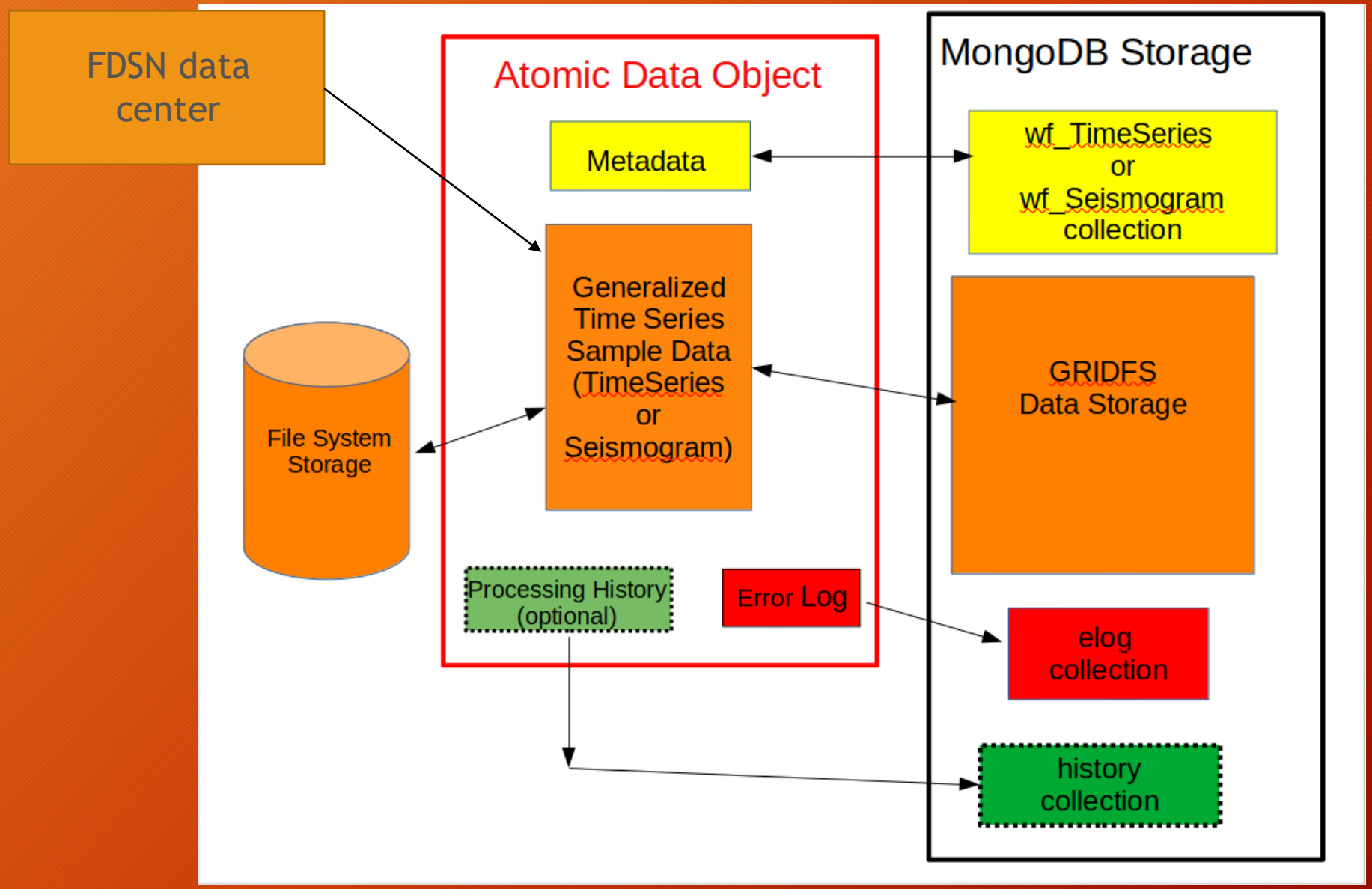
  MsPASS Extension
  - Db.save_data
  - Db.save_inventory
  - Db.save_catalog
- Read
  Stock MongoDB
  - Db.find_one
  - Db.find

  MsPASS Extension
  - Db.read_data
- Update
- Delete

Create: MsPASS save_data method concepts

- Key concepts:
  - Data object is torn to pieces with save
  - Different containers are saved in different “collections” or files
- Header (Metadata container) saved in
  - wf_miniseed
  - wf_TimeSeries
  - wf_Seismogram
- Sample data
  - Files
  - Gridfs (internal to MongoDB)
- Error log
- Processing History

Graphical Representation of Save/read operations



Database Queries

- A key concept of the R of CRUD
- You almost never want “all” the things a database manages
- All database systems use some “query language” to select the data desired
- Standard in RDBMS is “Structured Query Language” (SQL)
- MongoDB uses a different query language based on python dictionaries

MongoDB Query Language (MQL)

- Two query methods:
  - find_one – retrieve first document matching query
  - find – retrieve command cursor == list of documents matching query
  - arg0 of both is a python dictionary
- Many online sources
  - Most are poor
  - Most just echo examples in MongoDB documentation
- MsPASS advice
  - See User Manual section Using MongoDB with MsPASS
  - Today's tutorial and similar tutorial found here

How Queries are used in MsPASS

- Serial Atomic read loops

```
cursor = db.wf_miniseed.find(query)
for doc in cursor:
   d = db.read_data(doc,collection='wf_TimeSeries')
   ...  Processing functions …
   db.save_data(d)
```

- Serial Ensemble read loops

```
srcid = db.source.find(query).distinct('_id')
for sid in srcid:
   wf_query = {'source_id' : sid}
   cursor = db.wf_miniseed.find(wf_query)
   ensemble = db.read_data(cursor)
   … Processing functions
   db.save_data(ensemble,collection='wf_TimeSeries')
```